

S2

Discrete Random Variables

Learning Objectives

In this chapter, we will

- (a) define and identify discrete random variables and binomial variables;
 - (b) construct probability distribution relating to a given situation;
 - (c) calculate expectation and variance for discrete random variables;
 - (d) understand that the binomial distribution $B(n, p)$ is an example of probability distributions; and know the mean and variance of the distribution;
 - (e) use the binomial distribution to model practical situations and calculate probabilities using a graphic calculator.
- [Note: Students should know the conditions under which the binomial distribution is a suitable model. They will be asked to comment on the appropriate use of a model and the assumptions made.]

2.1 [# Introduction](#)

We can think of the term ‘variable’ as a name given to a quantity that can change. For instance, the salary of a worker in a certain factory varies across different workers. We say that the “salary of a worker in the factory” is a variable. The weights (or heights) of different members of a population can also be thought of as different values of the variable ‘weight (or height) of a member’.

Variables need to be measurable quantities, otherwise we cannot tell when and how they change, but they need not necessarily be of a numerical nature. For instance, ‘the weather’ at any particular time is a variable quantity, but we do not usually allocate numbers to measure the different kinds of weather. We usually use descriptions such as ‘rainy’, ‘fine’, ‘sunny’, and ‘cloudy’.

In this chapter we study variables that can be associated with statistical experiments and investigate some characteristics of their behaviours. Since these variables take values depending on randomness (chance), they are known as random variables. To give a mathematical definition of random variable, we first consider the sample space S of some experiment. A random variable is a function X , $X : S \rightarrow \mathbb{R}$, which assigns a specific real number x to each outcome in S .

For example, in the process of tossing two coins and we let X be the number of heads obtained. In this case, the sample space $S = \{HH, HT, TH, TT\}$ and X is a random variable, where $X(HH) = 2$, $X(HT) = 1$, $X(TH) = 1$ and $X(TT) = 0$.

In the example of rolling an ordinary die, the score rolled is a **discrete random variable** since it can only take 1, 2, 3, 4, 5 or 6, which is **countable**. In contrast, the random variable 'weight of a living human' is a **continuous random variable** since it can take any positive real number value.

More examples of discrete random variables include

- Number of heads obtained in tossing a coin 10 times;
- Number of girls in a family of 4 children;
- Number of red balls out of 3 balls drawn from a bag containing 2 red balls and 3 green balls.

2.2 # Probability Distribution

For each of the possible values that a discrete random variable can take, there is a probability associated with it. In the case of rolling an ordinary fair die, the probability of getting a 1, 2, 3, 4, 5 or 6 is $\frac{1}{6}$.

The convention is to use an uppercase letter, say X , as the random variable and the corresponding lowercase letter, say x , to represent one of the values it can take.

Let X be the score of the fair die. The table below lists all the possible values, x and their corresponding probabilities:

x	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

The above table is called the **probability distribution** of the random variable X . The probability distribution of a random variable tells us all the different values that the random variable can assume and their corresponding probabilities. The probability distribution can be in the form of a table, a list or it can be represented by a formula as follows:

$$P(X = x) = \frac{1}{6} \quad \text{for } x = 1, 2, 3, 4, 5, 6.$$

Each time the dice is thrown, each of the 6 values is equally likely to be observed. Here, X is said to follow a **discrete uniform distribution**.

For any probability distribution of **discrete** random variable, we have

1. $0 \leq P(X = x) \leq 1$ for all x ;
2. $\sum_{\text{all } x} P(X = x) = 1$.

Example 1

A fair coin is tossed 3 times in succession. Let X be the number of heads obtained. Tabulate the probability distribution of X .

Find

- (i) $P(X \geq 2)$,
- (ii) $P(0.8 \leq X \leq 3.2)$.

Solution

We shall list the set of all possible outcomes (recall from Chapter 1 that this is known as the sample space).

Sample Space = $\{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\}$.

HHT is one outcome where heads turn up in the first two tosses and tails in the third toss, and so when HHT occurs, we have $X = 2$. On the other hand, when we know $X = 2$, any one of the following could have occurred: HHT, HTH, THH .

X takes values 0, 1, 2 and 3 with the following probability distribution:

x	0	1	2	3
$P(X = x)$	$\left(\frac{1}{2}\right)^3 = \frac{1}{8}$	$3\left(\frac{1}{2}\right)^3 = \frac{3}{8}$	$3\left(\frac{1}{2}\right)^3 = \frac{3}{8}$	$\left(\frac{1}{2}\right)^3 = \frac{1}{8}$

$$\begin{aligned}
 \text{(i) } P(X \geq 2) &= P(X = 2) + P(X = 3) \\
 &= \frac{3}{8} + \frac{1}{8} \\
 &= \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } P(0.8 \leq X \leq 3.2) &= P(X = 1, 2, 3) \quad (\text{since } X \text{ takes discrete values only}) \\
 &= 1 - P(X = 0) \\
 &= \frac{7}{8}
 \end{aligned}$$

□

Example 2

The discrete random variable Y has probability distribution as follows:

y	-3	-2	-1	0	1
$P(Y = y)$	0.1	0.25	0.3	0.15	d

Find

- the value of d ,
- $P(Y > -1)$,
- $P\left(\left|Y + \frac{1}{2}\right| > 1\right)$,
- the mode of the distribution (i.e. the value of Y with the highest probability).

Solution

$$\begin{aligned} \text{(i)} \quad \sum_{\text{all } y} P(Y = y) &= 1 \quad \Rightarrow \quad 0.1 + 0.25 + 0.3 + 0.15 + d = 1 \\ &\Rightarrow \quad d = 0.2 \end{aligned}$$

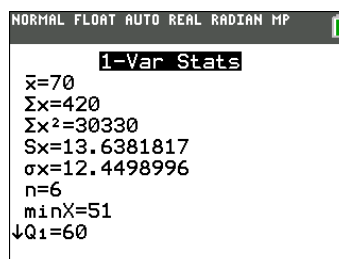
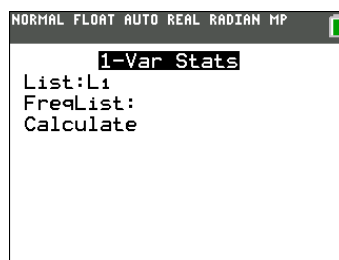
$$\begin{aligned} \text{(ii)} \quad P(Y > -1) &= P(Y = 0) + P(Y = 1) \\ &= 0.15 + 0.2 \\ &= 0.35 \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad P\left(\left|Y + \frac{1}{2}\right| > 1\right) &= P\left(Y + \frac{1}{2} > 1 \text{ or } Y + \frac{1}{2} < -1\right) \\ &= P\left(Y > \frac{1}{2} \text{ or } Y < -\frac{3}{2}\right) \\ &= P(Y = 1) + P(Y = -2) + P(Y = -3) \\ &= 0.2 + 0.25 + 0.1 \\ &= 0.55 \end{aligned}$$

- (iv) Since the probability of $Y = -1$ is the highest, the mode of the distribution is -1 . □

N-5

- ③ Press [2nd] [QUIT].
- ④ Press [STAT] > CALC.
- ⑤ Select **1:1-Var Stats**, press [ENTER].
Key in L₁ for List:, press [ENTER] to find $\bar{x}$.



□

Example 4

A fair die is thrown 60 times and the results are recorded in a frequency distribution table below:

score, x	1	2	3	4	5	6
frequency, f	9	11	12	8	13	7

Find the mean score.

Solution

To calculate the mean score, we multiply each score by its frequency, sum these values and then divide by 60 as follows:

$$\begin{aligned}\bar{x} &= \frac{\sum fx}{\sum f} \\ &= \frac{(1 \times 9) + (2 \times 11) + (3 \times 12) + (4 \times 8) + (5 \times 13) + (6 \times 7)}{60} = \frac{103}{30}\end{aligned}$$

or

$$\begin{aligned}\bar{x} &= \sum x \left(\frac{f}{m} \right) \text{ where } m = \sum f \\ &= \left(1 \times \frac{9}{60} \right) + \left(2 \times \frac{11}{60} \right) + \left(3 \times \frac{12}{60} \right) + \left(4 \times \frac{8}{60} \right) + \left(5 \times \frac{13}{60} \right) + \left(6 \times \frac{7}{60} \right) \\ &= \frac{103}{30}\end{aligned}$$

Using TI84+ to evaluate $\bar{x}$:

- ① Press **STAT**, select **1:Edit**, press **ENTER**.
- ② Go to L_1 and key in the x values. Go to L_2 and key in the f values.

[illegible]

- ③ Press [2nd] [QUIT].
- ④ Press [STAT] > CALC.
- ⑤ Select **1:1-Var Stats**, press [ENTER].
Key in L₁ for List: and L₂ for FreqList:, press [ENTER] to find $\bar{x}$.

NORMAL FLOAT AUTO REAL RADIAN MP

1-Var Stats

List:L1
FreqList:L2
Calculate

NORMAL FLOAT AUTO REAL RADIAN MP

1-Var Stats

$\bar{x}=3.433333333$
 $\Sigma x=206$
 $\Sigma x^2=866$
 $Sx=1.640242507$
 $\sigma x=1.626516387$
 $n=60$
 $\min X=1$
 $\downarrow Q_1=2$

Theoretical Approach

The probability distribution for the random variable X , the score on the fair die, is as shown:

[illegible]

We can obtain the expectation of X by multiplying each score by its corresponding probability and summing,

$$\begin{aligned}\text{Expectation} &= \sum_{\text{all } x} x P(X=x) \\ &= 1 \times \left(\frac{1}{6}\right) + 2 \times \left(\frac{1}{6}\right) + 3 \times \left(\frac{1}{6}\right) + 4 \times \left(\frac{1}{6}\right) + 5 \times \left(\frac{1}{6}\right) + 6 \times \left(\frac{1}{6}\right) \\ &= 3.5\end{aligned}$$

The expectation of a random variable X (or expected value of X) is what you would expect the average of the outcome of an experiment to be in the long run.

❧ Note

The expected value may not be a value taken by the discrete random variable. In Example 4, the expected value of X is 3.5 whereas the discrete random variable X takes integer values 1, 2, 3, 4, 5 or 6.

Note

If the probability distribution is symmetric about a particular value that is within the range of values that X can take, then expectation of X or $E(X)$ is equal to that particular value. In Example 5, the probability distribution is symmetric about 3.5, so $E(X) = 3.5$.

Example 6

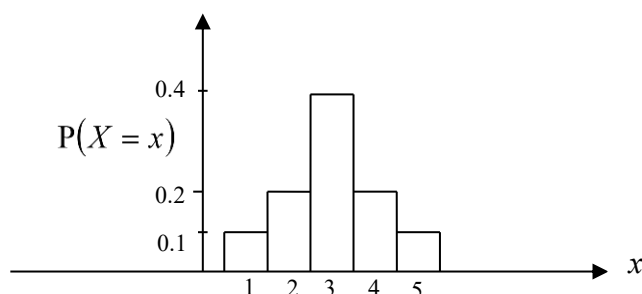
A random variable X has probability distribution as shown in the table below:

x	1	2	3	4	5
$P(X = x)$	0.1	0.2	0.4	0.2	0.1

Sketch the probability distribution of X and state $E(X)$.

Solution

The probability distribution is shown below as a bar chart.



Since the probability distribution is symmetric about the central value $x = 3$, so $E(X) = 3$. $\square$

2.5 Expectation of any function of Discrete Random Variable X , $E(g(X))$

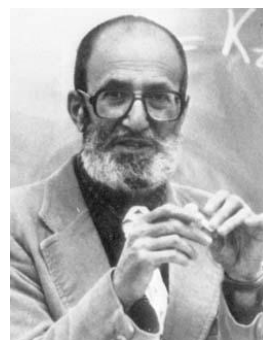
Let X be a discrete random variable with probability distribution $P(X = x)$.

The **expectation** of $g(X)$ is given by

$$E(g(X)) = \sum_{\text{all } x} g(x)P(X = x)$$

This result is so well-known that it is usually used without stating a name for it. Many people thought that $E(g(X)) = \sum_{\text{all } x} g(x)P(X = x)$ is a definition. This is in fact a

theorem which bears the name “The Law of the Unconscious Statistician”, first coined by Paul Halmos in the mid 1940s. This theorem is very useful, since it enables us to find $E(g(X))$ without first finding the probability distribution of $g(X)$.



Paul Halmos
(1912 – 2006)

In particular,

$$\begin{aligned} E(aX + b) &= \sum_{\text{all } x} (ax + b) P(X = x) \\ &= \sum_{\text{all } x} (ax P(X = x) + b P(X = x)) \\ &= a \sum_{\text{all } x} x P(X = x) + b \sum_{\text{all } x} P(X = x) \\ &= a E(X) + b, \text{ since } \sum_{\text{all } x} P(X = x) = 1. \end{aligned}$$

$E(aX + b) = a E(X) + b$ is a very useful result when calculating expectations of functions of X , where a and b are constants.

Useful Results

Given that a and b are constants,

1. $E(a) = a$,
2. $E(aX + b) = a E(X) + b$,
3. $E(aX \pm bY) = a E(X) \pm b E(Y)$, for any two random variables X and Y .

(2) has been proven above and (1) can be obtained from the proof of (2) by letting X be the random variable having only one possible value 0.

Example 7

A random variable X has the following probability distribution

x	1	2	3	4
$P(X = x)$	0.6	0.2	0.1	0.1

Calculate

- (i) $E(3X + 5)$,
- (ii) $E(3 - X^2)$.

Solution

First we calculate $E(X)$ and $E(X^2)$,

$$\begin{aligned} E(X) &= \sum_{\text{all } x} x P(X = x) \\ &= (1 \times 0.6) + (2 \times 0.2) + (3 \times 0.1) + (4 \times 0.1) \\ &= 1.7 \\ E(X^2) &= \sum_{\text{all } x} x^2 P(X = x) \\ &= (1^2 \times 0.6) + (2^2 \times 0.2) + (3^2 \times 0.1) + (4^2 \times 0.1) \\ &= 3.9 \end{aligned}$$

Using the above results,

$$\begin{aligned} \text{(i)} \quad E(3X + 5) &= 3E(X) + 5 \\ &= (3 \times 1.7) + 5 \\ &= 10.1 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad E(3 - X^2) &= E(3) - E(X^2) \\ &= 3 - 3.9 \\ &= -0.9 \end{aligned}$$

□

2.6 # Measures of Variation

In statistics, in order to describe the data set accurately, the statistician may like to know beyond the measures of central tendency, i.e. he may want to know the extent to which the data is scattered or spread out. For example, two or more classes may have the same mean mark for a test, and we would like to check for each class how close the marks are to the mean so as to understand the variation in ability of the class.

The **variance** of a data set is the average of the squared differences between each value and the mean.

Experimental Approach

The variance of n numbers $x_1, x_2, \dots, x_n$ taken from a population is known as the sample variance as we treat $x_1, x_2, \dots, x_n$ as a sample from that population (it is not the population itself). The sample variance, denoted by $S.V.$ is given as

$$S.V. = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Recall: $\bar{x}$ denotes the mean of the n numbers.

The **standard deviation** of a data set is the positive square root of the variance of the same data set, i.e. the standard deviation = $\sqrt{\text{variance}}$.

Let us look at two sets of data below:

5, 15, 20, 100 and 35, 35, 35, 35

They have the same mean 35 but the first set of data is more spread out.

$$\text{Its variance is } \frac{1}{n} \sum_{i=1}^4 (x_i - \bar{x})^2 = \frac{(5-35)^2 + (15-35)^2 + (20-35)^2 + (100-35)^2}{4} = 1437.5$$

which is greater than the variance of the second set of data, 0.

For data with frequency distribution, we can calculate the sample variance as follows:

$$S.V. = \frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{\sum_{i=1}^n f_i} \quad \text{which can be simplified to} \quad S.V. = \frac{\sum_{i=1}^n f_i x_i^2}{\sum_{i=1}^n f_i} - \bar{x}^2.$$

Recall: $\bar{x}$ denotes the mean of n numbers $x_1, x_2, \dots, x_n$ and $\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}$.

Proof:

$$S.V. = \frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{\sum_{i=1}^n f_i} = \frac{\sum_{i=1}^n f_i (x_i^2 - 2x_i \bar{x} + \bar{x}^2)}{\sum_{i=1}^n f_i} = \frac{\sum_{i=1}^n f_i x_i^2 - 2\bar{x} \sum_{i=1}^n f_i x_i + \bar{x}^2 \sum_{i=1}^n f_i}{\sum_{i=1}^n f_i} \text{ and since } \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i} = \bar{x},$$

$$S.V. = \frac{\sum_{i=1}^n f_i x_i^2}{\sum_{i=1}^n f_i} - 2\bar{x}^2 + \bar{x}^2 = \frac{\sum_{i=1}^n f_i x_i^2}{\sum_{i=1}^n f_i} - \bar{x}^2$$

We use the frequency distribution in Example 4 to calculate the sample variance of X :

Score, x	1	2	3	4	5	6
Frequency, f	9	11	12	8	13	7

$$S.V. = \frac{\sum_{\text{all } x} f(x - \bar{x})^2}{\sum f} \quad \text{where } \bar{x} = \frac{\sum f x}{\sum f} = \frac{206}{60} = \frac{103}{30}$$

$$= \frac{9 \times \left(1 - \frac{103}{30}\right)^2 + 11 \times \left(2 - \frac{103}{30}\right)^2 + \dots + 7 \times \left(6 - \frac{103}{30}\right)^2}{60}$$

$$= 2.645553 \approx 2.65.$$

or

$$S.V. = \sum_{\text{all } x} (x - \bar{x})^2 \frac{f}{m} \quad \text{where } m = \sum f$$

$$= \left(1 - \frac{103}{30}\right)^2 \times \frac{9}{60} + \left(2 - \frac{103}{30}\right)^2 \times \frac{11}{60} + \dots + \left(6 - \frac{103}{30}\right)^2 \times \frac{7}{60}$$

$$= 2.64555556 \approx 2.65.$$

or

$$S.V. = \frac{\sum f x^2}{m} - \bar{x}^2$$

$$= \frac{1^2 \times 9 + 2^2 \times 11 + \dots + 6^2 \times 7}{60} - \left(\frac{103}{30}\right)^2$$

$$= 2.64555556 \approx 2.65$$

2.7 Variance of a Discrete Random Variable

Let X be a discrete random variable, and let μ be the mean of X .

The **variance** of X denoted by $\text{Var}(X)$ or σ^2 is given by

$$\text{Var}(X) = E(X - \mu)^2 = E(X^2) - \mu^2.$$

Note that $\text{Var}(X) = E(X - \mu)^2$ can be simplified to $E(X^2) - \mu^2$ as follows :

$$\begin{aligned} \text{Var}(X) &= E(X - \mu)^2 \\ &= E(X^2 - 2X\mu + \mu^2) \\ &= E(X^2) - 2\mu E(X) + E(\mu^2) \\ &= E(X^2) - 2(\mu)^2 + \mu^2 \quad [\text{Note: } E(X) = \mu] \\ &= E(X^2) - \mu^2. \end{aligned}$$

The **standard deviation** of a discrete random variable X is the positive square root of the variance of X , i.e., the standard deviation = $\sqrt{\text{variance}}$ or σ .

Variance and standard deviation measure the spread of the values assumed by a random variable about its mean. A small variance or standard deviation indicates that most of the values are close to the mean.

Theoretical Approach

The probability distribution for the random variable X , the score on the die, is as shown below:

x	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

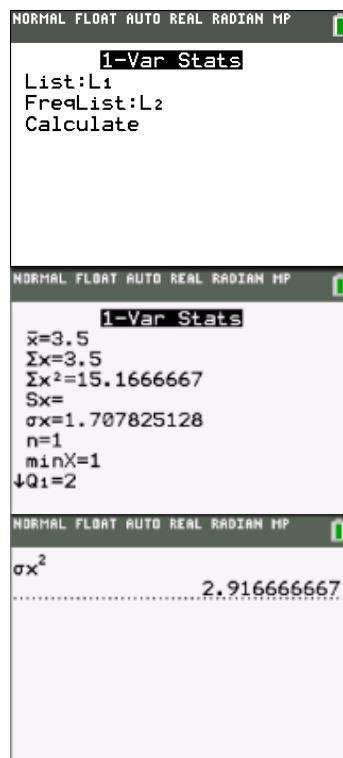
$$\begin{aligned} \text{Variance} &= E(X - \mu)^2 = \sum_{\text{all } x} (x - \mu)^2 P(X = x) \\ &= (1 - 3.5)^2 \times \left(\frac{1}{6}\right) + (2 - 3.5)^2 \times \left(\frac{1}{6}\right) + \dots + (6 - 3.5)^2 \times \left(\frac{1}{6}\right) \\ &= 2.92 \end{aligned}$$

Using TI84+ to evaluate Variance:

- ① Press **[STAT]**, select **1:Edit**, press **[ENTER]**.
- ② Go to L_1 and key in the x values. Go to L_2 and key in the $P(X = x)$ values.

L1	L2	L3	L4	L5	L6
1	.16667				
2	.16667				
3	.16667				
4	.16667				
5	.16667				
6	.16667				

- ③ Press $[2^{nd}][QUIT]$.
- ④ Press $[STAT] > CALC$.
- ⑤ Select **1:1-Var Stats**, press $[ENTER]$.
Key in L_1 for List: and L_2 for FreqList:, press $[ENTER]$ to find σ_x .
- ⑥ Press $[2^{nd}][QUIT]$.
- ⑦ Press $[VARS]$ and select 5:Statistics then 4: σ_x and press $[x^2]$.



By comparing $S.V. = \sum_{\text{all } x} (x - \bar{x})^2 \frac{f}{m}$ and $\text{Var}(X) = \sum_{\text{all } x} (x - \mu)^2 P(X = x)$, we notice the similarity in the definitions, where $\frac{f}{m}$ is the experimental probability and $P(X = x)$ is the theoretical probability.

Useful Results

Given that a and b are constants, it can be proven that

1. $\text{Var}(a) = 0$,
2. $\text{Var}(aX + b) = a^2 \text{Var}(X)$,
 $\text{Var}(aX \pm bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$
 if X and Y are **independent** random variables.

Example 8

A biased four sided die is thrown once. The score X has the following probability distribution:

x	1	2	3	4
$P(X = x)$	0.4	0.2	0.1	0.3

Find

- (i) $\text{Var}(X)$,
- (ii) $\text{Var}(3X)$,
- (iii) $\text{Var}(2)$,
- (iv) $\text{Var}(3X - 2)$.

Solution

$$E(X) = (1 \times 0.4) + (2 \times 0.2) + (3 \times 0.1) + (4 \times 0.3) = 2.3$$

$$E(X^2) = (1^2 \times 0.4) + (2^2 \times 0.2) + (3^2 \times 0.1) + (4^2 \times 0.3) = 6.9$$

$$\begin{aligned} \text{(i)} \quad \text{Var}(X) &= E(X^2) - \mu^2 \\ &= 6.9 - 2.3^2 = 1.61 \end{aligned}$$

Using the above results,

$$\text{(ii)} \quad \text{Var}(3X) = 3^2 \text{Var}(X) = 9 \times 1.61 = 14.49$$

$$\text{(iii)} \quad \text{Var}(2) = 0$$

$$\text{(iv)} \quad \text{Var}(3X - 2) = 3^2 \text{Var}(X) = 9 \times 1.61 = 14.49$$

□

Example 9

The random variables X and Y are independent and it is given that $E(X) = 2$, $\text{Var}(X) = 0.5$, $E(Y) = 5$ and $\text{Var}(Y) = 2$.

Find

$$\text{(i)} \quad E(4X - 3Y),$$

$$\text{(ii)} \quad \text{Var}(4X - 3Y).$$

Solution

$$\text{(i)} \quad E(4X - 3Y) = 4E(X) - 3E(Y) = 4(2) - 3(5) = -7$$

$$\begin{aligned} \text{(ii)} \quad \text{Var}(4X - 3Y) &= (4)^2 \text{Var}(X) + (-3)^2 \text{Var}(Y) \quad \text{since } X \text{ and } Y \text{ are independent} \\ &= (4)^2 (0.5) + (-3)^2 (2) \\ &= 26 \end{aligned}$$

□

Note

If X_1 and X_2 are two independent observations of X , then

$$E(X_1) = E(X_2) = E(X) = \mu \quad \text{and} \quad \text{Var}(X_1) = \text{Var}(X_2) = \text{Var}(X) = \sigma^2$$

In general, for n independent observations from the distribution of X ,

$$\begin{aligned} E(X_1 + X_2 + \cdots + X_n) &= nE(X) = n\mu, \\ \text{Var}(X_1 + X_2 + \cdots + X_n) &= n\text{Var}(X) = n\sigma^2. \end{aligned}$$

Example 10

A biased die is thrown once. Let X be the score shown on the die. X has the following probability distribution:

x	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{2}{3}$

- Let S be twice the score shown on the die. Find the expectation and variance of S .
- The die is now thrown twice. Let T be the sum of the two scores obtained. Find the expectation and variance of T .

Solution

$$E(X) = \sum_{\text{all } x} x P(X = x) = 5$$

$$E(X^2) = \sum_{\text{all } x} x^2 P(X = x) = 27\frac{2}{3}$$

$$\text{Var}(X) = E(X^2) - \mu^2 = 27\frac{2}{3} - (5)^2 = \frac{8}{3}$$

$$(i) \quad E(S) = E(2X) = 2E(X) = 2 \times 5 = 10.$$

$$\begin{aligned} \text{Var}(S) &= \text{Var}(2X) \\ &= (2)^2 \text{Var}(X) \\ &= 4 \times \left(\frac{8}{3}\right) \\ &= 10\frac{2}{3}. \end{aligned}$$

$$(ii) \quad T = X_1 + X_2, \text{ where the two scores, } X_1 \text{ and } X_2 \text{ are independent observations of } X.$$

$$\begin{aligned} E(T) &= E(X_1 + X_2) \\ &= E(X_1) + E(X_2) = 2E(X) = 2(5) = 10. \end{aligned}$$

$$\begin{aligned} \text{Var}(T) &= \text{Var}(X_1 + X_2) \\ &= \text{Var}(X_1) + \text{Var}(X_2) \\ &= 2 \text{Var}(X) \\ &= 2 \times \left(\frac{8}{3}\right) = \frac{16}{3} = 5\frac{1}{3}. \end{aligned}$$

□

Note

- Note that $X_1 + X_2 \neq 2X$. $X_1 + X_2$ is said to be the sum of two independent observations of X , while $2X$ is said to be two times an observation of X .
- Interestingly, they have the same mean. $E(X_1 + X_2) = 2E(X) = E(2X)$.
- However, they have different variances. If X_1 and X_2 are independent $\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2) = 2 \times \text{Var}(X)$, but $\text{Var}(2X) = 2^2 \text{Var}(X) = 4 \times \text{Var}(X)$.
- $\text{Var}(2X) > \text{Var}(X_1 + X_2)$ which indicates that the values of $2X$ is more spread out about the mean as compared to those of $X_1 + X_2$.

2.8 # Bernoulli Trials

In a soccer match, the referee tosses a coin once at the beginning of the match to determine which one of the two teams kicks off the match. Tossing the coin once to determine which team kicks off the match is an example of a **Bernoulli trial**, named after a Swiss mathematician Jacob Bernoulli.

In statistics, a Bernoulli trial refers to a **single** experiment which has only two possible outcomes: 'success' and 'failure'. Another example of a Bernoulli trial that we often come across is rolling a die in which obtaining a six is considered a 'success' and other numbers is considered a 'failure'.

(Refer to Appendix A to find out more about the Bernoulli random variable)

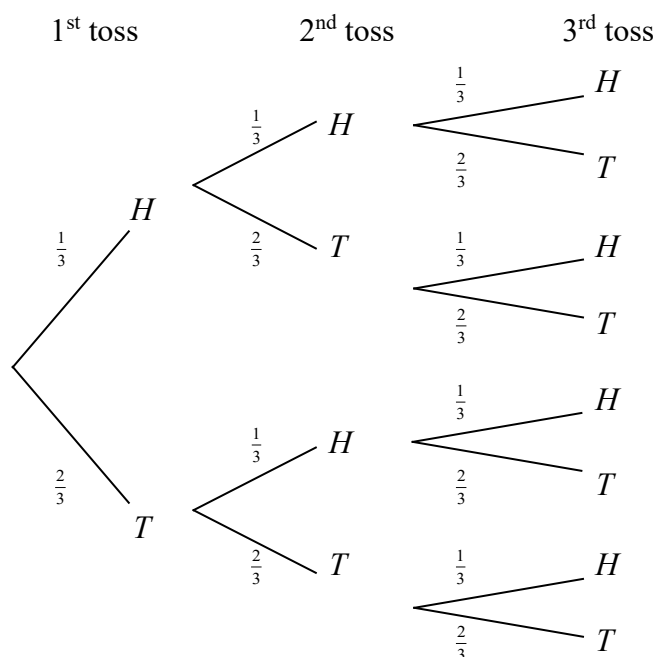


Jacob Bernoulli
(1654 – 1705)

2.9 Binomial Distribution

Let us consider tossing a biased coin where the probability of obtaining a head in any toss of the coin is $\frac{1}{3}$. For any toss of the coin, let H be the outcome of obtaining a head, and T be the outcome of obtaining a tail.

If the coin is tossed thrice independently (three independent Bernoulli trials), we can create a probability tree diagram to list all possible outcomes.



We will abbreviate $P(H \cap H \cap T)$ as $P(HHT)$.

What is the probability that exactly two heads are obtained in three tosses of the coin?

$P(\text{obtaining exactly two heads out of 3 tosses})$

$$= P(HHT) + P(HTH) + P(THH)$$

$$= \left(\frac{1}{3}\right)\left(\frac{1}{3}\right)\left(\frac{2}{3}\right) + \left(\frac{1}{3}\right)\left(\frac{2}{3}\right)\left(\frac{1}{3}\right) + \left(\frac{2}{3}\right)\left(\frac{1}{3}\right)\left(\frac{1}{3}\right) \quad (\text{from tree diagram})$$

$$= 3\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right).$$

Notice that in each game, there are three ways where we can obtain two heads (two ‘successes’) and one tail (one ‘failure’). This is the same as saying that of the three tosses, we must choose two tosses that will obtain heads.

So, $P(\text{obtaining exactly two heads out of 3 tosses})$

$$\begin{aligned} &= \binom{3}{2} \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right) \quad (\text{choosing 2 ‘successes’ from 3 tosses}) \\ &= 3 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right) \end{aligned}$$

In general, let X be the number of heads obtained in n tosses of a coin (n Bernoulli trials). If the n tosses are performed independently, and the probability of obtaining a head in each toss remains a constant p , then the probability distribution of X is given by

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n,$$

where

- $\binom{n}{x}$ or nC_x is the number of ways of choosing x objects from n distinct objects;
- p is the probability of landing on a head (a ‘success’); and
- $(1-p)$ is the probability of landing on a tail (a ‘failure’).

2.10 # Characteristics of a Binomial Random Variable

A Bernoulli experiment that is repeated n times, such as tossing of a coin n times in the example above, are called **binomial experiments**.

The random variable, X , which denotes the number of ‘successes’, is called the **binomial random variable**.

The probability distribution of X is called the **binomial distribution**.

Some examples of a binomial random variable include:

- (a) The number of heads obtained when a coin is tossed 10 times.
- (b) The number of sixes obtained when a die is thrown 5 times.
- (c) The number of ace cards obtained when 5 cards are drawn **with** replacement from a deck of 52 playing cards.

The random variables below do not follow a binomial distribution:

- (d) The number on the top face of a die that is tossed once.
- (e) The number of tosses of a die until a six is first obtained.
- (f) The number of green cards drawn from a bag that contains 10 cards: 3 green cards and 7 blue cards. A card is drawn at random five times in succession **without** replacement.

From the examples above, we notice that a binomial random variable has the following characteristics:

1. The experiment consists of n (fixed) repeated **independent** trials.
2. Each trial has **only two outcomes**: a ‘success’ or a ‘failure’.
3. The probability of a ‘success’, denoted by p , is **constant** in each trial.

2.11 Notation of the Binomial Distribution

The number of trials, n , and the probability of success, p , are called the parameters of the binomial distribution. We write $X \sim B(n, p)$ to denote that X follows a binomial distribution with parameters n and p .

The probability of obtaining x successes in n trials is given by

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \text{ where } x = 0, 1, 2, 3, \dots, n.$$

The formula for $P(X = x)$ can be found in MF26.

Example 11

Suppose we roll an ordinary die 3 times. Let X be the number of sixes obtained.

- Find $P(X = 2)$.
- Tabulate the probability distribution of X .
- Find $P(X \leq 2)$.

Solution

- Let $X \sim B\left(3, \frac{1}{6}\right)$.

Method 1: Using Formula

$$\begin{aligned} P(X = 2) &= \binom{3}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^1 \\ &= \frac{5}{72} \end{aligned}$$

Method 2: Using TI84+

$$\begin{aligned} P(X = 2) &= 0.069444 \\ &= 0.0694 \text{ (to 3 s.f.)} \end{aligned}$$

- Press $\boxed{2\text{nd}} \boxed{[\text{DISTR}]}$.
- Select **A: binompdf**, press $\boxed{[\text{ENTER}]}$.
- Key in number of trials n , probability of success in one trial p , followed by the X value and press $\boxed{[\text{ENTER}]}$.

```
binompdf
trials:3
p:1/6
x value:2
Paste

binomPdf(3,1/6,2)
.....0694444444
```

- Method 1: Using Formula

x	0	1	2	3
$P(X = x)$	$\left(\frac{5}{6}\right)^3 = \frac{125}{216}$	${}^3C_1 \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^2 = \frac{25}{72}$	${}^3C_2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^1 = \frac{5}{72}$	$\left(\frac{1}{6}\right)^3 = \frac{1}{216}$

Method 2: Using TI84+

- ① Press $\boxed{Y=}$. In Y_1 , press $\boxed{2nd}$ $\boxed{DISTR}$, select **A:binompdf** , press $\boxed{ENTER}$.
- ② Key in 3, 1/6, and press $\boxed{X,T,\theta,n}$.

binompdf				
trials:3				
p:1/6				
x value:X				
Paste				

X	Y1			
0	.5787			
1	.34722			
2	.06944			
3	.00463			
4	0			
5	0			
6	0			
7	0			
8	0			
9	0			
10	0			

X=0

- ③ Press $\boxed{2nd}$ $\boxed{TABLE}$.

(iii) Method 1: Using Table in (ii)

$$\begin{aligned}
 P(X \leq 2) &= P(X = 0) + P(X = 1) + P(X = 2) \\
 &= 0.99537 \\
 &= 0.995 \text{ (to 3 s.f.)}
 \end{aligned}$$

Method 2: Using TI84+

$$\begin{aligned}
 P(X \leq 2) &= 0.99537 \\
 &= 0.995 \text{ (to 3 s.f.)}
 \end{aligned}$$

- ① Press $\boxed{2nd}$ $\boxed{DISTR}$, select **B: binomcdf** , press $\boxed{ENTER}$.
- ② Key in number of trials n , probability of success in one trial p , followed by the X value and press $\boxed{ENTER}$

binomcdf				
trials:3				
p:1/6				
x value:2				
Paste				

binomcdf(3,1/6,2)				
.9953703704				

Note

1. For discrete random variable, the function $P(X \leq x)$ is known as the **cumulative distribution function** of X .

$$P(X \leq x) = P(X = 0) + P(X = 1) + P(X = 2) + \dots + P(X = x)$$

If $X \sim B(n, p)$, it follows from the above that $\sum_{x=0}^n P(X = x) = P(X \leq n) = 1$.

2. The binomial distribution is closely related to the binomial expansion. If $X \sim B(n, p)$, we notice that $P(X = r)$ is the $(r + 1)^{\text{th}}$ term in the binomial expansion

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + \underbrace{\binom{n}{r} a^{n-r} b^r}_{(r+1)^{\text{th}} \text{ term}} + \dots + b^n$$

with $a = 1 - p$ and $b = p$.

Example 12

Meiyi takes a test consisting of 20 multiple choice questions. Each question has four options with only one option as the correct answer. Meiyi has no knowledge of the correct answer to any of the questions and she decides to randomly pick one of the four options as her answer to each of the 20 questions. Find the probability that she will get

- (a) at most 11 questions correct,
- (b) at least five questions correct,
- (c) at least 2 but less than 14 questions correct.

Solution

Let X be the number of questions (out of 20) answered correctly.

Then $X \sim B\left(20, \frac{1}{4}\right)$.

(a) $P(X \leq 11) = 0.999065 = 0.999$ (to 3 s.f.)

Using TI84+:

- ① Press **2nd** [DISTR], select **B: binomcdf**, press **ENTER**.
- ② Key in 20, 1/4, 11 and press **ENTER**.

```
binomcdf
trials:20
p:1/4
x value:11
Paste
```

(b) $P(X \geq 5) = 1 - P(X \leq 4)$
 $= 0.58516$
 $= 0.585$ (to 3 s.f.)

```
1-binomcdf(20,1/4,4)
.....5851584992
```

(c) $P(2 \leq X < 14) = P(2 \leq X \leq 13)$
 $= P(X \leq 13) - P(X \leq 1)$
 $= 0.97566$
 $= 0.976$ (to 3 s.f.)

```
binomcdf(20,1/4,
13)-binomcdf(20,
1/4,1)
.9756578634
```

□

Example 13

Referring to Example 12, if the probability of Meiyi correctly answering at least r questions in the test of 20 multiple choice questions is at least 0.7, find the greatest value of r .

Solution

Let X be the number of questions (out of 20) answered correctly.

Then, $X \sim B\left(20, \frac{1}{4}\right)$.

Given $P(X \geq r) \geq 0.7$

$$1 - P(X \leq r-1) \geq 0.7$$

$$P(X \leq r-1) \leq 0.3$$

From table,

$$P(X \leq 4-1) = P(X \leq 3) = 0.22516 < 0.3$$

$$P(X \leq 5-1) = P(X \leq 4) = 0.41484 > 0.3$$

$\therefore$ greatest $r = 4$.

X	Y1			
0	0			
1	.00317			
2	.02431			
3	.09126			
4	.22516			
5	.41484			
6	.61717			
7	.78578			
8	.89819			
9	.95907			
10	.98614			

X=0

Using TI84+:

- ① In Y_1 , press $\boxed{2\text{nd}} \boxed{[\text{DISTR}]}$, select **B:**
binomcdf , press $\boxed{\text{ENTER}}$.
- ② Key in 20, 0.25, $X-1$.
- ③ Press $\boxed{2\text{nd}} \boxed{[\text{TBLSET}]}$ for table setup and key in the settings as shown on the right.
- ④ Press $\boxed{2\text{nd}} \boxed{[\text{TABLE}]}$.

binomcdf	TABLE SETUP
trials:20	TblStart=0
p:1/4	ΔTbl=1
x value:X-1	IndPnt: Auto Ask
Paste	Depend: Auto Ask

X	Y1
0	.00317
1	.02431
2	.09126
3	.22516
4	.41994
5	.61717
6	.78578
7	.93819
8	.95907
9	.98614
10	

X=0

Example 14

In a particular nation, $p\%$, where $p > 50$, of its citizens are of blood group A. A random sample of 30 citizens is selected. It is given that the probability of no less than ten but at most fifteen citizens in the random sample are of blood group A is 0.3. Find the value of p .

Solution

Let X be the number of citizens (out of 30) who are of blood group A.

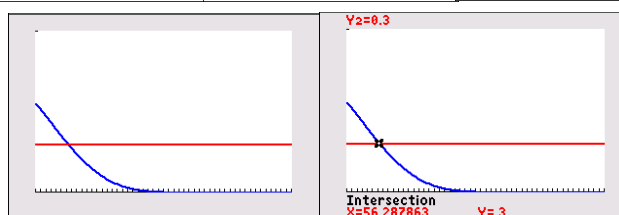
$$\text{Then } X \sim B\left(30, \frac{p}{100}\right).$$

$$\text{Given } P(10 \leq X \leq 15) = 0.3$$

$$P(X \leq 15) - P(X \leq 9) = 0.3$$

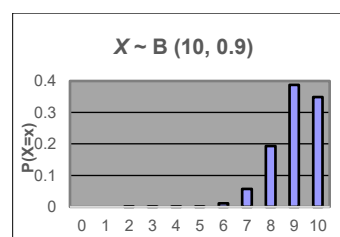
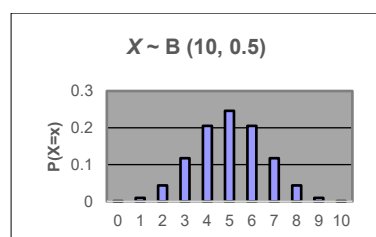
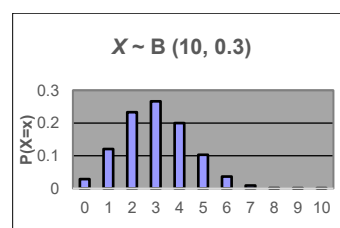
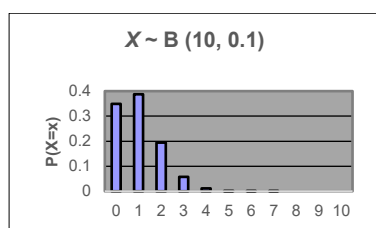
$$\begin{aligned} \text{From graph, } p &= 56.288 \\ &= 56.3 \quad (\text{to 3 s.f.}) \end{aligned}$$

binomcdf	Plot1 Plot2 Plot3	WINDOW
trials:30	Y1 $B \leftarrow \text{binomcdf}(30, X/100, 9)$	Xmin=50
p: X/100	Y2 $B0.3$	Xmax=100
x value:15	Y3=	Xscl=1
Paste	Y4=	Ymin=0
	Y5=	Ymax=1
	Y6=	Yscl=1
	Y7=	Xres=1
	Y8=	ΔX=.18939393939394
	Y9=	TraceStep=.378787878788



2.12 # Diagrammatic Representation of the Binomial Distribution

Let us consider a binomial random variable X , where $X \sim B(10, p)$. The probability distributions of X for different values of p are illustrated as follow:



Note

The distribution becomes more symmetrical when p is close to 0.5.

2.13 Expectation and Variance of the Binomial Distribution

$$\text{If } X \sim B(n, p), \text{ then } E(X) = np \text{ and } \text{Var}(X) = np(1-p).$$

The formulae for $E(X)$ and $\text{Var}(X)$ can be found in MF27.

Refer to Appendix A for proof.

Example 15

The random variable X has the binomial distribution with mean 2 and variance $\frac{24}{13}$.

Find $P(X = 4)$.

Solution

$$E(X) = np = 2$$

$$\text{Var}(X) = np(1-p) = \frac{24}{13}$$

Solving for n and p , we have $p = \frac{1}{13}$ and $n = 26$.

$$\text{Hence } X \sim B\left(26, \frac{1}{13}\right).$$

$$\therefore P(X = 4) = 0.0900$$

2.13 Mode of the Binomial Distribution

The **mode** of X is the value of X with the highest probability of occurrence. In general, there are two ways to find the mode of the binomial distribution.

Method 1:

Consider the probability distribution of X and the mode is the value of X with the highest probability.

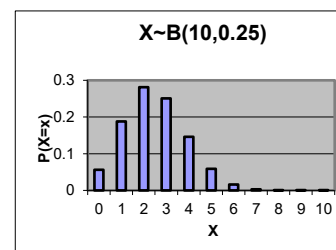
In general, the mode is near the mean. So, in finding the mode, we can consider only the probabilities of the values of X that are near to the mean.

Consider $X \sim B(10, 0.25)$.

The mean is $np = 2.5$ and the mode is 2 as shown in the figure on the right.

X	Y1			
0	.05631			
1	.18771			
2	.28157			
3	.25028			
4	.146			
5	.0584			
6	.01622			
7	.00309			
8	3.9E-4			
9	2.9E-5			
10	9.5E-7			

X=0



Method 2: By Recurrence Formula

Refer to Appendix B for proof.

Note

The probability distribution of a binomial random variable is typically unimodal, while in some cases two adjacent values can have the same mode. The probability decreases as the value of X tends away from the mode.

Example 16

Age-related macular degeneration (AMD) is an eye disease that usually affects the macula in older adults. The incidence of AMD is estimated at 19 per 100 adults who are between 66 to 74 years of age. On a randomly chosen weekday afternoon, 40 patients who are between 66 to 74 years of age are waiting to see the eye specialist, what is the most likely number of patients suffering from AMD?

Solution

Let A be the number of patients (out of 40) who are suffering from AMD.

Then $A \sim B(40, 0.19)$.

Note that $E(A) = 7.6$.

From table,

$$P(A = 6) = 0.13969$$

$$P(A = 7) = 0.15915$$

$$P(A = 8) = 0.15399$$

Hence the most likely number of patients suffering from AMD is 7.

X	Y1			
0	2.2E-4			
1	.00205			
2	.00938			
3	.02786			
4	.06045			
5	.10209			
6	.13969			
7	.15915			
8	.15399			
9	.12843			
10	.09339			
X=7				

Using TI84+:

- ① In Y_1 , press $\boxed{2nd}$ [DISTR], select **A:binompdf** , press $\boxed{ENTER}$.
- ② Key in 40, 0.19, X.
- ③ Press $\boxed{2nd}$ [TBLSET] for table setup and key in the settings as shown on the right.
- ④ Press $\boxed{2nd}$ [TABLE].

```
binompdf
trials:40
p:0.19
x value:X
Paste
```

```
TABLE SETUP
TblStart=0
ΔTbl=1
Indent: Auto Ask
Depend: Auto Ask
```

Note

Refer to Appendix B on the use of the recurrence formula to find the mode.

Summary

- A **discrete random variable** is a countable numerical variable whose value depends on randomness (chance).
- A **discrete probability distribution** can be in the form of a table, a list or it can be represented by a formula displaying all possible outcomes together with the probability of each outcome.
- The mean of a data set is often denoted by $\bar{x}$ and is given as

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{or} \quad \bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}, \text{ where } f_i \text{ is the frequency of } x_i.$$

- The expectation or mean of a discrete random variable X is
- The expectation or mean, of a function or sum of functions of a discrete random variable X is

$$E(X) = \sum_{\text{all } x} x P(X = x).$$

$$E(g(X)) = \sum_{\text{all } x} g(x) P(X = x).$$

$$E(f(X) + g(X)) = E(f(X)) + E(g(X)).$$

- Some useful results for expectation (or expected value):

$$E(a) = a;$$

$$E(aX + b) = a E(X) + b;$$

$$E(aX \pm bY) = a E(X) \pm b E(Y);$$

where a and b are constants, and X and Y are two random variables (not necessarily independent).

- The sample variance of a data set is denoted as $S.V.$ and is given by

$$S.V. = \frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{\sum_{i=1}^n f_i} \quad \text{or} \quad S.V. = \frac{\sum_{i=1}^n f_i x_i^2}{\sum_{i=1}^n f_i} - \bar{x}^2.$$

where f_i is the frequency of x_i .

- The variance of a discrete random variable X is denoted as $\text{Var}(X)$ or σ^2 , where

$$\text{Var}(X) = E(X - \mu)^2.$$

It can also be expressed as

$$\text{Var}(X) = E(X^2) - \mu^2.$$

- The standard deviation of a discrete random variable, X is the positive square root of the variance of X , i.e., the standard deviation $= \sqrt{\text{variance}}$ or σ .
- Given that a and b are constants, it can be proven that
 1. $\text{Var}(a) = 0$,
 2. $\text{Var}(aX + b) = a^2 \text{Var}(X)$,
 3. $\text{Var}(aX \pm bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$
 if X and Y are **independent** random variables.

Binomial distribution

If $X \sim B(n, p)$, then

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n$$

$$E(X) = np, \text{Var}(X) = np(1-p).$$

Appendix A – Expectation and Variance of the Binomial Distribution

Let X_i be a Bernoulli random variable with probability p for success and probability $(1 - p)$ for failure. Then $X_i \sim B(1, p)$.

We let X_i be 1 if the i^{th} Bernoulli trial is successful and 0 if otherwise.

x	0	1
$P(X_i = x)$	$1 - p$	p

$$\text{Then } E(X_i) = 1(p) + 0(1 - p) = p.$$

Now we let X be the number of successes in n independent Bernoulli trials with probability p for success in each trial.

Since $X = X_1 + X_2 + \dots + X_n \sim B(n, p)$, then

$$E(X) = E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n) = np. \quad \square$$

To find the variance of a binomial random variable, we can sum the variances of n Bernoulli random variables similar to the approach used above. But first, we need to find $E(X_i^2)$.

$$E(X_i^2) = 1^2(p) + 0^2(1 - p) = p.$$

$$\text{Since } \text{Var}(X) = E(X^2) - [E(X)]^2,$$

$$\text{Var}(X_i) = E(X_i^2) - [E(X_i)]^2 = p - p^2.$$

Since $X_1, X_2, \dots, X_n$ are independent, then

$$\begin{aligned} \text{Var}(X) &= \text{Var}(X_1 + X_2 + \dots + X_n) \\ &= \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n) \\ &= n(p - p^2) \\ &= np(1 - p). \end{aligned}$$

□

Appendix B - Recurrence Formula to Find the Mode of a Binomial Distribution

Another way to find the mode is by considering the values of x such that $P(X = k) < P(X = k + 1)$ and those x such that $P(X = k) > P(X = k + 1)$. The recurrence formula is given by $\frac{P(X = k + 1)}{P(X = k)} = \frac{(n - k)p}{(k + 1)(1 - p)}$ where $k = 0, 1, 2, \dots, (n - 1)$.

Proof for $\frac{P(X = k + 1)}{P(X = k)} = \frac{(n - k)p}{(k + 1)(1 - p)}$:

$$\begin{aligned}\frac{P(X = k + 1)}{P(X = k)} &= \frac{\binom{n}{k + 1} p^{k + 1} (1 - p)^{n - k - 1}}{\binom{n}{k} p^k (1 - p)^{n - k}} \\ &= \frac{(n - k)p}{(k + 1)(1 - p)}\end{aligned}$$

□

Revisit Example 16

Case 1:

Consider $P(A = k + 1) > P(A = k)$:

Since $\frac{P(X = k + 1)}{P(X = k)} = \frac{(n - k)p}{(k + 1)(1 - p)}$, we have $(n - k)p > (k + 1)(1 - p)$ (*).

By substituting $n = 40$ and $p = 0.19$ into (*), we have

$(40 - k) \times 0.19 > (k + 1) \times 0.81 \Rightarrow k < 6.79$, i.e., the largest value of k is 6.

Recall that $P(A = k + 1) > P(A = k)$.

Hence when $k = 6$, we have $P(A = 7) > P(A = 6)$.

When $k = 1$, we have $P(A = 2) > P(A = 1)$ and so on.

Thus, we have $P(A = 7) > P(A = 6) > \dots > P(A = 1) > P(A = 0)$.

Case 2:

Consider $P(A = k + 1) < P(A = k)$:

Since $\frac{P(X = k + 1)}{P(X = k)} = \frac{(n - k)p}{(k + 1)(1 - p)}$, we have $(n - k)p < (k + 1)(1 - p)$.

Similarly, $(40 - k) \times 0.19 < (k + 1) \times 0.81 \Rightarrow k > 6.79$, i.e., the smallest value of k is 7.

Recall that $P(A = k + 1) < P(A = k)$.

Hence when $k = 7$, we have $P(A = 7) > P(A = 8)$. When $k = 8$, we have $P(A = 8) > P(A = 9)$ and so on.

Thus, we have $P(A = 7) > P(A = 8) > P(A = 9) > \dots$

Now, we observe that $P(A = 7)$ has the highest probability in both Case 1 and Case 2.

Therefore, the mode is 7. □

Tutorial 2: Discrete Random Variables

1. A discrete random variable X takes all non-negative integer values with probabilities given by $P(X = x) = k(4^{-x})$. Find k .

2. **DHS Prelim 9758/2023/02/Q5**
 Tim plays a game with 3 outcomes having the score of 2, 5 and 6. The corresponding probabilities are k , $\frac{2}{7}$ and $\frac{4}{7}$ respectively, where k is a constant.
 - (a) State the value of k . [1]

Tim pays \$ m in total to play the game twice. The outcome of each game is independent. He receives an amount corresponding to twice the absolute difference of the two scores from each game.

 - (b) If Tim is expected make a profit, determine the range of values of m . [3]
 - (c) For $m = 2.3$, explain why Tim may not profit for every 2 such games that he plays. [1]

3. **N2020/P2/Q5**
 Shania and Tina are playing a game. Shania has a bag containing one green disc, r red discs and $2r$ blue discs, where $r > 1$. A red disc is worth 5 points, a blue disc is worth 2 points and the green disc is worth 0 points. Tina takes two discs from the bag at random. Tina's score is found by multiplying together the number of points for each of the two discs she takes.
 - (i) State Tina's possible scores. [1]
 - (ii) Show that the expectation of Tina's score is $\frac{27r-11}{3r+1}$, and find an expression for the variance. [7]
 - (iii) Given that the variance of Tina's score is 38, find the value of r . [2]

4. A newsagent sells a particular magazine. The number of copies that he can sell in a month has the probability distribution shown in the table. Each month he sells them at \$1.20 per copy, making a profit of 50 cents on each magazine sold. Any remaining unsold at the end of the month are wasted and valueless.

Number	10	11	12	13	14
Probability	0.15	0.25	0.30	0.2	0.1

Find

- (i) the mean number that he could sell if he had sufficient copies;
- (ii) the mean monthly profit if he always bought 11 copies.

5. **IJC17/BT2/7 (modified)**

The discrete random variable X has probability function

$$P(X = x) = \begin{cases} k\left(\frac{5}{2} - x\right), & x = 0, 2, \\ k(x - 2), & x = 3, \\ 0, & \text{otherwise,} \end{cases}$$

where k is a positive constant.

- (i) Show that $k = \frac{1}{4}$. [1]
- (ii) Find $E(X)$ and show that $\text{Var}(X) = \frac{7}{4}$. [3]
- (iii) A discrete random variable Y has $E(Y) = 2$ and $\text{Var}(Y) = \frac{24}{5}$. Find the values of $E(2X - 3Y)$ and $\text{Var}(2X - 3Y)$, stating any assumption necessary for the calculations to be valid. [5]

Given that X_1 and X_2 are two independent observations of X ,

- (iv) find the probability that the sum of X_1 and X_2 exceeds 4, given that the sum is at least 3. [3]

6. An archer took 15 shots at a target during each training session and counted the number of hits. When the archer repeated the training session a large number of times, the mean number of hits was noted to be 3. Calculate
- (i) the probability of hitting the target with a single shot,
 - (ii) the variance of the number of hits in a single training session.

7. (a) 40% of chocolates made by Amazing Chocolates are truffles and the rest are caramel. A random sample of 30 pieces of chocolates is collected for quality control. Find the probability that
- (i) the sample has exactly 4 pieces of truffles,
 - (ii) the sample has fewer than 8 pieces of truffles,
 - (iii) the 15th piece of chocolate is the 3rd truffle in the sample.
- (b) A box of 50 pieces of Bazing Chocolates contains 20 pieces of truffles and 30 pieces of caramel. In a party of 12 children, everyone takes one piece of chocolate randomly out of the box. Find the probability that exactly 4 children get a truffle each.
8. **N2017/P2/Q5**
 A bag contains 6 red counters and 3 yellow counters. In a game, Lee removes counters at random from the bag, one at a time, until he has taken out 2 red counters. The total number of counters Lee removes from the bag is denoted by T .
- (i) Find $P(T = t)$ for all possible values of t . [3]
 - (ii) Find $E(T)$ and $\text{Var}(T)$. [2]
- Lee plays this game 15 times.
- (iii) Find the probability that Lee has to take at least 4 counters out of the bag in at least 5 of his 15 games. [2]
9. **N2012/P2/Q9 (modified)**
 In an opinion poll before an election, a sample of 30 voters is obtained. The number of voters in the sample who support the Alliance Party is denoted by A . State in context, two conditions that must be assumed for A to be modelled by a Binomial Distribution. [2]
- (a) It is given that 5 % of the voters in the election support the Alliance Party.
 - (i) What is the probability that fewer than 6 of the 30 voters support the Alliance Party? [2]
 - (ii) What will be the most likely number of voters who support the Alliance Party in a sample of 40 voters. [2]
 - (iii) It is given that, in a sample of 50 voters, the probability that at least m voters who support the Alliance Party is less than 0.4. Find the least possible value of m .
 - (iv) If a sample of n number of voters is obtained and the probability of getting at least a supporter is not more than 0.75. Find the maximum value of n . [3]
 - (b) In the next election, another sample of 30 voters is obtained and the percentage of the voters is p where $p > 50$. It is given that the probability of getting 15 to 20 supporters is 0.3. Find the value of p . [4]

10. **N2017/P2/Q9**

On average 8% of a certain brand of kitchen lights are faulty. The lights are sold in boxes of 12.

- (i) State, in context, two assumptions needed for the number of faulty lights in a box to be well modelled by a binomial distribution. [2]

Assume now that the number of faulty lights in a box has a binomial distribution.

- (ii) Find the probability that a box of 12 of these kitchen lights contains at least 1 faulty light. [1]

The boxes are packed into cartons. Each carton contains 20 boxes.

- (iii) Find the probability that each box in one randomly selected carton contains at least one faulty light. [1]
 (iv) Find the probability that there are at least 20 faulty lights in a randomly selected carton. [2]
 (v) Explain why the answer to part (iv) is greater than the answer to part (iii). [1]

The manufacturer introduces a quick test to check if lights are faulty. Lights identified as faulty are discarded. If a light is faulty there is a 95% chance that the quick test will correctly identify the light as faulty. If the light is not faulty, there is a 6% chance that the quick test will incorrectly identify the light as faulty.

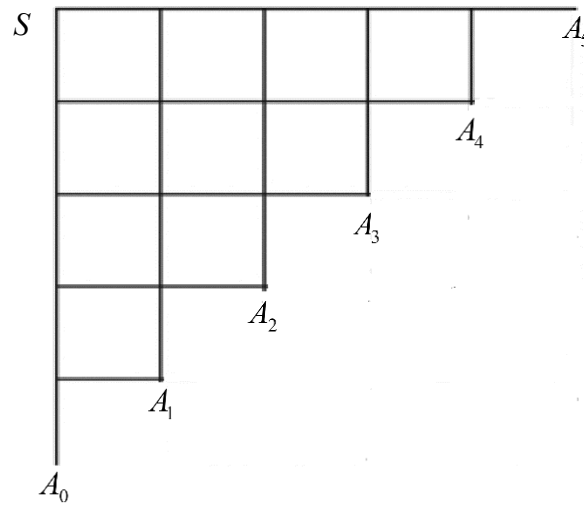
- (vi) Find the probability that a light identified as faulty by the quick test is **not** faulty. [3]
 (vii) Find the probability that the quick test correctly identifies lights as faulty or not faulty. [1]
 (viii) Discuss briefly whether the quick test is worthwhile. [1]

11. **NJC19/Prelim/II/5 (modified) *Core Skill 6/10***

The random variable X is the number of successes in n independent trials of an experiment in which the probability of a success at any one trial is p .

- (i) Show that $\frac{P(X = k + 1)}{P(X = k)} = \frac{(n - k)p}{(k + 1)(1 - p)}$, where $k = 0, 1, 2, \dots, (n - 1)$. [2]
 (ii) Using the result in part (i), find the most probable number of successes when $n = 99$ and $p = \frac{7}{9}$. [3]

12. MI19/Prelim/II/10



A particle moves one step each time either to the right or downwards through a network of connected paths as shown above. The particle starts at S , and, at each junction, randomly moves one step to the right with probability p , or one step downwards with probability q , where $q = 1 - p$. The steps taken at each junction are independent. The particle finishes its journey at one of the 6 points labelled A_i , where $i = 0, 1, 2, 3, 4, 5$ (see diagram). Let $\{X = i\}$ be the event that the particle arrives at point A_i .

- (i) Show that $P(X = 2) = 10p^2q^3$. [2]
- (ii) After experimenting, it is found that the particle will end up at point A_2 most of the time. By considering the mode of X or otherwise, show that $\frac{1}{3} < p < \frac{1}{2}$. [4]

The above setup is a part of a two-stage computer game.

- If the particle lands on A_0 , the game ends immediately and the player will not win any points.
- If the particle lands on A_i , where $i = 2$ or 4 , then the player gains 2 points.
- If the particle lands on A_i , where $i = 1, 3$ or 5 , then the player proceeds on to the next stage, where there is a probability of 0.4 of winning the stage. If he wins the stage, he gains 5 points; otherwise he gains 3 points.

Let Y be the number of points gained by the player when one game is played.

- (iii) If $p = 0.4$, determine the probability distribution of Y . [4]
- (iv) Hence find the expectation and variance of Y . [2]

Answers

1. $\frac{3}{4}$
2. (a) $\frac{1}{7}$ (b) $0 < m < 2.45$
3. (i) 0, 4, 10, 25 (ii) $\frac{360r^2 + 300r - 340}{(3r+1)^2}$ (iii) 3
4. (i) 11.85 (ii) \$5.32
5. (ii) 1 (iii) $-4; \frac{251}{5}$ (iv) $\frac{8}{29}$
6. (i) 0.2 (ii) 2.4
7. (ai) 0.00120 (ii) 0.0435 (iii) 0.0127 (b) 0.234
8. (ii) $E(T) = 2.86$ $\text{Var}(T) = 0.765$ (iii) 0.238
9. (a)(i) 0.997 (a)(ii) 2 (a)(iii) 4 (a)(iv) 27 (b) 72.4
10. (ii) 0.632 (iii) 0.000104 (iv) 0.458 (vi) 0.421 (vii) 0.9408
11. 77
12. (iv) 2.74; 1.86

Skill Sets

Statistics Chapter 2: Discrete Random Variables

No.	Skills	Examples of questions involving the skills																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
1.	List down the possible outcomes of an event and identify the possible values a random variable can attain. Construct the probability distribution of a random variable.	Example A fair coin is tossed 3 times in succession. Let X be the number of heads obtained. Tabulate the probability distribution of X. Solution We shall list the set of all possible outcomes (recall from Chapter 1 that this is known as the sample space). Sample Space = $\{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\}$. HHT is one outcome where heads turn up in the first two tosses and tails in the third toss, and so when HHT occurs, we have $X = 2$. On the other hand, when we know $X = 2$, any one of the following could have occurred: HHT, HTH, THH. X takes values 0, 1, 2 and 3 with the following probability distribution: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>$P(X = x)$</td><td>$\left(\frac{1}{2}\right)^3 = \frac{1}{8}$</td><td>$3\left(\frac{1}{2}\right)^3 = \frac{3}{8}$</td><td>$3\left(\frac{1}{2}\right)^3 = \frac{3}{8}$</td><td>$\left(\frac{1}{2}\right)^3 = \frac{1}{8}$</td></tr></table>	x	0	1	2	3	$P(X = x)$	$\left(\frac{1}{2}\right)^3 = \frac{1}{8}$	$3\left(\frac{1}{2}\right)^3 = \frac{3}{8}$	$3\left(\frac{1}{2}\right)^3 = \frac{3}{8}$	$\left(\frac{1}{2}\right)^3 = \frac{1}{8}$																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
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2.	Finding mean and variance of a data set by algebraic method or using GC	[Lecture Notes Example 3] Find the mean of: 80 51 72 89 68 60. Solution $\bar{x} = \frac{1}{n} \sum_{i=1}^6 x_i = \frac{80 + 51 + 72 + 89 + 68 + 60}{6} = 70$ Using TI84+ to find $\bar{x}$ and variance: - Press [STAT], select 1:Edit, press [ENTER]. - Go to L_1 and key in the x values. <table><tr><th>L_1</th><th>L_2</th><th>L_3</th><th>L_4</th><th>L_5</th><th>L_6</th></tr><tr><td>80</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>51</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>72</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>89</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>68</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>60</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> - Press [2nd] [QUIT]. - Press [STAT] > CALC. <table><tr><th>L_1</th><th>L_2</th><th>L_3</th><th>L_4</th><th>L_5</th><th>L_6</th></tr><tr><td>80</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>51</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>72</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>89</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>68</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>60</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> - Select 1:1-Var Stats, press [ENTER]. 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- ⑥ To find variance of this given data set:
Press [2nd] [QUIT] to return to home screen and press [VAR] followed by [5] followed by [4] [x²] [ENTER]

NORMAL FLOAT AUTO REAL RADIAN HP	NORMAL FLOAT AUTO REAL RADIAN HP	NORMAL FLOAT AUTO REAL RADIAN HP
VAR Y-VARS COLOR 1:Window... 2:Zoom... 3:GDB... 4:Picture & Background... 5:Statistics... 6:Table... 7:String...	XY Σ EQ TEST PTS 1:n 2:x̄ 3:Sx 4:σx 5:y 6:Sy 7:σy 8:minX 9↓maxX	σx ²155

[Lecture Notes Example 4]

A fair die is thrown 60 times and the results are recorded in a frequency distribution table below:

score, x	1	2	3	4	5	6
frequency, f	9	11	12	8	13	7

Find the mean score.

Solution

To calculate the mean score, we multiply each score by its frequency, sum these values and then divide by 60 as follows:

$$\bar{x} = \frac{\sum fx}{\sum f}$$

$$= \frac{(1 \times 9) + (2 \times 11) + (3 \times 12) + (4 \times 8) + (5 \times 13) + (6 \times 7)}{60} = \frac{103}{30}$$

or

$$\bar{x} = \sum x \left(\frac{f}{m} \right) \text{ where } m = \sum f$$

$$= \left(1 \times \frac{9}{60} \right) + \left(2 \times \frac{11}{60} \right) + \left(3 \times \frac{12}{60} \right) + \left(4 \times \frac{8}{60} \right) + \left(5 \times \frac{13}{60} \right) + \left(6 \times \frac{7}{60} \right) = \frac{103}{30}$$

Using TI84+ to find $\bar{x}$ and variance:

- Press [STAT], select **1:Edit**, press [ENTER].
- Go to L₁ and key in the x values.
Go to L₂ and key in the f values.

L1	L2	L3	L4	L5	2
1	9				
2	11				
3	12				
4	8				
5	13				
6	7				

- Press [2nd] [QUIT].
- Press [STAT] > CALC.
- Select **1:1-Var Stats**, press [ENTER].

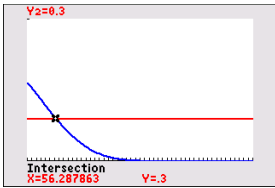
Key in L₁ for List: and L₂ for FreqList:, press [ENTER] to find $\bar{x}$.

NORMAL FLOAT AUTO REAL RADIAN HP	NORMAL FLOAT AUTO REAL RADIAN HP
1-Var Stats List:L1 FreqList:L2 Calculate	1-Var Stats x̄=3.433333333 Σx=206 Σx ² =866 Sx=1.640242507 σx=1.626516387 n=60 minX=1 ↓Q1=2

		⑥ To find variance of this given data set: Press [2nd] [QUIT] to return to home screen and press [VARS] followed by [5] followed by [4] [x²] [ENTER] NORMAL FLOAT AUTO REAL RADIAN HP VARS Y-VARS COLOR 1:Window... 2:Zoom... 3:GDB... 4:Picture & Background... 5:Statistics... 6:Table... 7:String... NORMAL FLOAT AUTO REAL RADIAN HP XV Σ EQ TEST PTS 1:n 2:x 3:Σx 4:Σx² 5:ȳ 6:Sy 7:σy 8:minX 9:maxX NORMAL FLOAT AUTO REAL RADIAN HP σx²2.645555556														
3.	Calculate mean and variance from probability distribution table	[Lecture Notes Example 10] A biased die is thrown once. Let X be the score shown on the die. X has the following probability distribution: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>$P(X = x)$</td><td>$\frac{1}{15}$</td><td>$\frac{1}{15}$</td><td>$\frac{1}{15}$</td><td>$\frac{1}{15}$</td><td>$\frac{1}{15}$</td><td>$\frac{2}{3}$</td></tr></table> (i) Let S be twice the score shown on the die. Find the expectation and variance of S. Solution $E(X) = \sum_{\text{all } x} x P(X = x) = 5$ $E(X^2) = \sum_{\text{all } x} x^2 P(X = x) = 27\frac{2}{3}$ $\text{Var}(X) = E(X^2) - [E(X)]^2 = 27\frac{2}{3} - (5)^2 = \frac{8}{3}$	x	1	2	3	4	5	6	$P(X = x)$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{2}{3}$
x	1	2	3	4	5	6										
$P(X = x)$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{2}{3}$										
4.	With the given $E(X)$, $E(Y)$, $\text{Var}(X)$ and $\text{Var}(Y)$, where X and Y are independent discrete random variables, evaluate expectation and variance of $aX \pm bY$ using properties of expectation and variance.	Example The discrete random variable X has probability function $P(X = x) = \begin{cases} \frac{1}{4} \left(\frac{5}{2} - x \right), & x = 0, 2, \\ \frac{1}{4} (x - 2), & x = 3, \\ 0, & \text{otherwise,} \end{cases}$ It is found that $E(X) = 1$ and $\text{Var}(X) = \frac{7}{4}$. A discrete random variable Y has $E(Y) = 2$ and $\text{Var}(Y) = \frac{24}{5}$. Find the values of $E(2X - 3Y)$ and $\text{Var}(2X - 3Y)$, stating any assumption necessary for the calculations to be valid. [5] $E(2X - 3Y) = 2E(X) - 3E(Y) = 2(1) - 3(2) = -4$														

		$\begin{aligned}\text{Var}(2X - 3Y) &= 2^2 \text{Var}(X) + 3^2 \text{Var}(Y) \\ &= 4\left(\frac{7}{4}\right) + 9\left(\frac{24}{5}\right) \\ &= \frac{251}{5}\end{aligned}$ X and Y must be independent.
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5.	Identify the parameters of binomial distribution Use binompdf and binomcdf appropriately.	[Lecture Notes Example 12] Meiyi takes a test consisting of 20 multiple choice questions. Each question has four options with only one option as the correct answer. Meiyi has no knowledge of the correct answer to any of the questions and she decides to randomly pick one of the four options as her answer to each of the 20 questions. Find the probability that she will get (a) at most 11 questions correct, (b) at least five questions correct, (c) at least 2 but less than 14 questions correct. Solution Let X be the number of questions (out of 20) answered correctly. Then $X \sim B\left(20, \frac{1}{4}\right)$. (a) $P(X \leq 11) = 0.999065 = 0.999$ (to 3 s.f.) Using TI84+: ① Press [2nd] [DISTR], select B: binomcdf, press [ENTER]. ② Key in 20, 1/4, 11 and press [ENTER]. <pre>binomcdf trials:20 p:1/4 x value:11 Paste</pre> (b) $P(X \geq 5) = 1 - P(X \leq 4)$ <pre>1-binomcdf(20,1/4,4)5851584992</pre> $= 0.58516$ $= 0.585$ (to 3 s.f.) (c) $P(2 \leq X < 14)$ $= P(2 \leq X \leq 13)$ $= P(X \leq 13) - P(X \leq 1)$ <pre>binomcdf(20,1/4,13)-binomcdf(20,1/4,1)9756578634</pre>
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6.	Using table in GC to find unknowns such as n or r in binomial distribution. Note: Table method only works for integer values	[Lecture Notes Example 13] Referring to point 5 Example above, if the probability of Meiyl correctly answering at least r questions in the test of 20 multiple choice questions is at least 0.7, find the greatest value of r. Solution Let X be the number of questions (out of 20) answered correctly. Then, $X \sim B\left(20, \frac{1}{4}\right)$. Given $P(X \geq r) \geq 0.7$ $1 - P(X \leq r-1) \geq 0.7$ $P(X \leq r-1) \leq 0.3$ From table, $P(X \leq 4-1) = P(X \leq 3) = 0.22516 < 0.3$ $P(X \leq 5-1) = P(X \leq 4) = 0.41484 > 0.3$ $\therefore$ greatest $r = 4$. <table border="1"><thead><tr><th>X</th><th>Y1</th></tr></thead><tbody><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>.00317</td></tr><tr><td>2</td><td>.02431</td></tr><tr><td>3</td><td>.09126</td></tr><tr><td>4</td><td>.22516</td></tr><tr><td>5</td><td>.41484</td></tr><tr><td>6</td><td>.51717</td></tr><tr><td>7</td><td>.59878</td></tr><tr><td>8</td><td>.69819</td></tr><tr><td>9</td><td>.79907</td></tr><tr><td>10</td><td>.88614</td></tr></tbody></table> X=0	X	Y1	0	0	1	.00317	2	.02431	3	.09126	4	.22516	5	.41484	6	.51717	7	.59878	8	.69819	9	.79907	10	.88614
X	Y1																									
0	0																									
1	.00317																									
2	.02431																									
3	.09126																									
4	.22516																									
5	.41484																									
6	.51717																									
7	.59878																									
8	.69819																									
9	.79907																									
10	.88614																									
7.	Use graph to solve when probability of success is unknown	[Lecture Notes Example 14] In a particular nation, $p\%$, where $p > 50$, of its citizens are of blood group A. A random sample of 30 citizens is selected. It is given that the probability of no less than ten but at most fifteen citizens in the random sample are of blood group A is 0.3. Find the value of p. Solution Let X be the number of citizens (out of 30) who are of blood group A. Then $X \sim B\left(30, \frac{p}{100}\right)$. Given $P(10 \leq X \leq 15) = 0.3$ $P(X \leq 15) - P(X \leq 9) = 0.3$ From graph, $p = 56.288$ $= 56.3$ (to 3 s.f.) 																								

8. Find mean, variance (using formula) and mode (using GC) of a given random variable that is modelled by a binomial distribution.

[Lecture Notes Example 16]

Age-related macular degeneration (AMD) is an eye disease that usually affects the macula in older adults. The incidence of AMD is estimated at 19 per 100 adults who are between 66 to 74 years of age. On a randomly chosen weekday afternoon, 40 patients who are between 66 to 74 years of age are waiting to see the eye specialist, what is the most likely number of patients suffering from AMD?

Solution

Let A be the number of patients (out of 40) who are suffering from AMD.

Then $A \sim B(40, 0.19)$.

Note that

$$E(A) = np = 40(0.19) = 7.6$$

X	Y_1			
0	2.2E-4			
1	.00205			
2	.00938			
3	.02786			
4	.06045			
5	.10209			
6	.13969			
7	.15915			
8	.15399			
9	.12843			
10	.09339			
X=7				

$$\text{Var}(A) = np(1-p) = 40(0.19)(0.81) = 6.156.$$

From table,

$$P(A = 6) = 0.13969$$

$$P(A = 7) = 0.15915$$

$$P(A = 8) = 0.15399$$

Hence the most likely number of patients suffering from AMD is 7.

Using TI84+:

- ① In Y_1 , press **[2nd]** **[DISTR]**, select **A:binompdf(**, press **[ENTER]**.

- ② Key in 40, 0.19, X.

```
binompdf
trials:40
p:0.19
x value:X
Paste
```

- ③ Press **[2nd]** **[TBLSET]** for table setup and key in the settings as shown on the right.

```
TABLE SETUP
TblStart=0
ΔTbl=1
IndPnt: Auto Ask
Depend: Auto Ask
```

- ④ Press **[2nd]** **[TABLE]**.

9.	Using the *formula of the probability distribution function of a binomial random variable to solve questions and to find the mode *this is given in MF27.	Example The random variable X is the number of successes in n independent trials of an experiment in which the probability of a success at any one trial is p. (i) Show that $\frac{P(X = k + 1)}{P(X = k)} = \frac{(n - k)p}{(k + 1)(1 - p)}$, where $k = 0, 1, 2, \dots, (n - 1)$. [2] (ii) Using the result in part (i), find the most probable number of successes when $n = 99$ and $p = \frac{7}{9}$. [3] Solution: (i) $\begin{aligned} \frac{P(X = k + 1)}{P(X = k)} &= \frac{\binom{n}{k + 1} p^{k + 1} (1 - p)^{n - k - 1}}{\binom{n}{k} p^k (1 - p)^{n - k}} \\ &= \frac{\frac{n!}{(k + 1)!(n - k - 1)!} p^{k + 1} (1 - p)^{n - k - 1}}{\frac{n!}{k!(n - k)!} p^k (1 - p)^{n - k}} \\ &= \frac{1}{(k + 1)} \frac{p}{(1 - p)} \\ &= \frac{(n - k)p}{(k + 1)(1 - p)} \end{aligned}$ Let k be the most probable number $P(X = k + 1) < P(X = k)$ and $P(X = k - 1) < P(X = k)$ Consider $P(X = k + 1) < P(X = k)$. $\frac{P(X = k + 1)}{P(X = k)} < 1$ $\frac{\frac{7}{9}(99 - k)}{(k + 1)\left(1 - \frac{7}{9}\right)} < 1$
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	$77 - \frac{7}{9}k < \frac{2}{9}k + \frac{2}{9}$ $k > 76\frac{7}{9} \quad (1)$ i.e. $P(X = 77) > P(X = 78) > P(X = 79) > \dots$ $P(X = k) > P(X = k - 1)$ $\frac{P(X = k)}{P(X = k - 1)} > 1$ $\frac{\frac{7}{9}(99 - k + 1)}{(k)\left(1 - \frac{7}{9}\right)} > 1$ $k < 77\frac{7}{9} \quad (2)$ i.e. $P(X = 77) > P(X = 76) > P(X = 75) > \dots$ Hence most probable number of successes is 77.
10	You should also use the summary provided in the chapter to help you revise through the topic.
11	The learning objectives at the first page of the lecture notes is good for knowing the important concepts that are taught in this chapter.

Supplementary Exercise 2: Discrete Random Variables

1. HCI/Prelim/2017/II/Q6

A biased tetrahedral (4-sided) die has its faces numbered '-1', '0', '2' and '3'. It is thrown onto a table and the random variable X denotes the number on the face in contact with the table. The probability distribution of X is as shown.

x	-1	0	2	3
$P(X = x)$	$\frac{1}{8}$	$\frac{1}{2}$	$\frac{1}{8}$	$\frac{1}{4}$

- (i) The random variable Y is defined by $X_1 + X_2$, where X_1 and X_2 are 2 independent observations of X . Show that $P(Y = 2) = \frac{3}{16}$. [2]
- (ii) In a game, a player pays \$2 to throw two such biased tetrahedral dice simultaneously on a table. For each die, the number on the face in contact with the table is the score of the die. The player receives \$16 if the maximum of the two scores is -1, and receives \$3 if the sum of the two scores is prime. For all other cases, the player receives nothing. Find the player's expected gain in the game. [4]

2. DHS/Prelim/2018/II/Q6

A test consists of 15 multiple choice questions, where each question has n possible options, of which only one is correct. A student took the test by randomly choosing an answer to each question. It is known that the probability of answering exactly 3 questions correctly is the same as the probability of answering exactly 4 questions correctly.

- (i) By forming an equation in terms of n , find the value of n . [3]
Each correct answer is awarded 3 marks and each incorrect answer carries a penalty of 1 mark. The score is the total marks awarded based on the number of correct and incorrect answers.
- (ii) Find the expected score, s , obtained by the student. [3]
- (iii) Find the probability that the score obtained by the student is within 4 marks of s . [2]

3. **TPJC/Prelim/2017/II/Q5**

An unbiased six-sided die is rolled twice. The random variable X represents the higher of the two values if they are different, and their common value if they are the same. The probability distribution of X is given by the formula

$$P(X = r) = k(2r - 1) \quad \text{for } r = 1, 2, 3, 4, 5, 6.$$

(i) Find the exact value of k , giving your answer as a fraction in its simplest form. [2]

(ii) Find the expectation of X . [2]

A round of the game consists of rolling the unbiased six-sided die twice, and X is taken as the score for the round. A player plays three rounds of the game.

(iii) Find the probability that the total score for the three rounds is 16. [2]

4. **TPJC/Prelim/2018/II/Q5**

An unbiased die has three faces painted red, two faces painted green and one face painted blue. A red face has a score of 1 point, a green face has a score of 2 points and a blue face has a score of 3 points.

Two such dice are thrown and the sum of their scores is denoted by X .

(i) Show that $P(X = 4) = \frac{5}{18}$ and find the probability distribution of X . [3]

(ii) Find $E(X)$ and show that $\text{Var}(X) = \frac{10}{9}$. [2]

Suppose now a red face has a score of 3 points, a green face has a score of 2 points, and a blue face has a score of 1 point.

(iii) Deduce the expectation and variance of the sum of scores obtained from a throw of two such dice. [2]

5. **IJC/Prelim/2017/II/Q8**

At a hospital, records show that 84.5% of patients turn up for their appointments. It is known that on any day, the doctor has time to see 20 patients.

On one particular day, there are 20 patients who make appointments to see the doctor.

- (i) State, in this context, one condition that must be met for the number of patients who turn up for their appointments to be well modelled by a binomial distribution.

[1]

For the remainder of this question, assume that the condition stated in part (i) is met.

- (ii) Find the probability that more than 15 patients turn up for their appointments. [2]

- (iii) Given that at least 12 patients turn up for their appointments, find the probability that more than 2 patients fail to turn up for their appointments. [3]

- (iv) To improve efficiency, the hospital decides to increase the number of appointments that can be made on each day. Given that there will still be enough time for the doctor to see 20 patients, find the greatest number of appointments that can be made so that there is a probability of at least 0.85 of the doctor having time to see all patients who turn up. [2]

6. **EJC/Prelim/2018/II/Q7**

When Mr. Lee sends a text message to any of his students over the weekend, he gets a reply, on average, about 6 out of 10 times. On a particular weekend, Mr. Lee sends a text message to 25 students.

- (i) State, in the context of this question, two assumptions needed to model the number of students that reply by a binomial distribution. [2]

- (ii) Explain why one of the assumptions stated in part (i) may not hold in this context. [1]

Assume now that these assumptions do in fact hold. Stating the distribution clearly,

- (iii) find the probability that at least half of the students reply. [3]

It is given instead that the probability of a randomly chosen student replying is p . Find the least value of p such that there is at least a 90% chance of more than 20 students replying to the text message. [3]

7. **JJC/Prelim/2018/II/Q8**

A jar contains 10 blue and 8 red marbles. Five marbles are randomly drawn from the box, one by one and without replacement.

(i) Explain why it is inappropriate to model the number of blue marbles by a binomial distribution. [1]

(ii) Find the probability that exactly three marbles are blue. [3]

Another jar contains 20 blue and 12 red marbles. n marbles are randomly drawn from the box, one by one and with replacement. The number of red marbles drawn is denoted by R .

(iii) Given that the mean of R is 4.5, find n and $P(R > 4)$. [3]

(iv) Given instead that $P(R = 0 \text{ or } 1) < 0.01$, write down an inequality for n and find the least value of n . [3]

8. **IJC/Prelim/2018/II/Q5**

The random variable X has the distribution $B(25, p)$, where $0 < p < 1$. Given that $P(X \leq 1) = 0.15$, write down an equation for the value of p and find this value numerically. Hence find $\text{Var}(X)$. [4]

9. **PJC/2018/BT2/II/8**

Tomatoes are sold in boxes of 16. On average, 15% of them are damaged.

(i) State, in context, two assumptions needed for the number of damaged tomatoes in a box to be well modelled by a binomial distribution. [2]

Assume now that the number of damaged tomatoes in a box has a binomial distribution.

(ii) Find the most likely number of damaged tomatoes in a randomly chosen box. [2]

(iii) Find the probability that a box contains less than 6 damaged tomatoes. [1]

(iv) A box contains less than 6 damaged tomatoes. Find the probability that it contains at least 2 damaged tomatoes. [3]

(v) Two boxes of tomatoes are randomly selected. Find the probability that one box has at least 2 damaged tomatoes and the other box has none. [2]

(vi) A market trader buys 80 boxes of tomatoes. Find the probability that there are more than 60 boxes with at least 2 damaged tomatoes, giving your answers to 4 decimal places. [3]

10. **TJC/2018/BT2/II/8**

Robin participates in a game show with 2 preliminary rounds and 1 final round. To proceed to the final round, she needs to obtain a positive total score from the two preliminary rounds.

In the first preliminary round, Robin needs to answer 2 questions, each having 5 possible answers. Robin chooses an answer randomly for each question. For each question she answers correctly, she will score 50 points. However, she will lose any points she has scored if she answers the second question wrongly.

- (i) Show that Robin's expected score in the first round is 12. [4]

In the second preliminary round, Robin needs to answer 3 questions, each also having 5 possible answers. Robin chooses an answer randomly for each question. For each question she answers correctly, she will score 12 points. For each question she answers wrongly, she will lose k points.

- (ii) Find the largest possible integer value of k if Robin is expected to proceed to the final round. [5]

11. **ACJC/Prelim/2017/II/Q9**

- (i) A procedure for accepting or rejecting a large batch of manufactured articles is such that an inspector first selects and examines a random sample of 10 articles from the batch. If the sample contains at least 2 defective articles, the batch is rejected.

It is known that the proportion of articles that are defective is 0.065. Show that the probability that a batch of articles is accepted is 0.866, correct to three significant figures. [1]

To confirm the decision, another inspector follows the same procedure with another random sample of 10 articles from the batch. If the conclusion of both inspectors are the same, the batch will be accepted or rejected as the case may be. Otherwise, one of the inspectors will select a further random sample of 10 from the same batch to examine. The batch is then rejected if there are at least 2 defective articles. Otherwise, it is accepted. Find

- (a) the probability that a batch is eventually accepted, [3]
 (b) the expected number of articles examined per batch. [4]

- (ii) In order to cut labour cost, an alternative procedure is introduced. A random sample of 10 articles is taken from the batch and if the sample contains not more than 1 defective article then the batch is accepted. If the sample contains more than 2 defective articles, the batch is rejected. If the sample contains exactly 2 defective articles, a second sample of 10 articles is taken and if this contains no defective article then the batch is accepted. Otherwise, the batch is rejected. Given that the proportion of defective articles in the batch is p , show that the probability that the batch is accepted is A where

$$A = (1 + 9p)(1 - p)^9 + 45p^2(1 - p)^{18}. \quad [2]$$

If the probability that, of 100 batches inspected, more than 80 of them will be accepted is 0.98, find the value of p . [3]

12. **RI/Prelim/2017/II/Q7**

An unbiased cubical die has the number 1 on one face, the number 2 on two faces and the number 3 on three faces. Adrian invites Benny to play a game. In each round, Benny rolls the die twice. Adrian pays Benny \$ a if the total score is 2 and \$3 if the total score is 3. However, if the total score is 4, Benny pays Adrian \$2. No payment is made otherwise.

- (i) Find the probability that Adrian pays Benny at least 5 times in 20 rounds. [4]

The random variable X represents Benny's winnings in each round.

- (ii) Given that $a = 6$, find the probability distribution of X . Hence, help Benny decide if he should accept Adrian's invitation to play the game. Justify your answer. [5]

- (iii) Determine the value of a for the game to be fair. [1]

13. **SRJC/Prelim/2017/II/Q5**

A random variable X has the probability distribution given in the following table.

x	2	3	4	5
$P(X = x)$	0.2	a	b	0.45

Given that $E(|X - 4|) = \frac{11}{10}$, find the values of a and b . [3]

Two independent observations of X are taken. Find the probability that one of them is 2 and the other is at most 4. [2]

14. **2017/BT2/DHS/P2/Q5 (modified)**

The probability distribution function of a discrete random variable R is defined by

$$P(R = r) = \begin{cases} \frac{1}{r(r+1)}, & \text{for all positive integer values of } r, \\ 0, & \text{otherwise.} \end{cases}$$

It is given that $P(R \leq n) = 1 - \frac{1}{n+1}$, where n is a positive integer.

(i) Find $\lim_{n \rightarrow \infty} P(R \leq n)$, showing your working clearly. [1]

(ii) Find $P(2 \leq R \leq 10)$. [2]

15. **2017/BT2/NYJC/P2/Q7**

Two discrete random variables X and Y are related by $Y = a|1 - X|$, where a is a constant. The probability distribution of X is given by

$$P(X = x) = kx^2 \quad \text{for } x = -3, -1, 1 \text{ and } 3,$$

where k is a positive real constant.

(i) Find k . [2]

(ii) Given that $E(Y) = 5.6$, find a and show that $\text{Var}(Y) = 5.44$. [4]

16. **2017/BT2/MJC/P1/Q9**

An airline company sold 105 tickets for a particular flight. Based on past experience, 8% of passengers do not turn up for the flight. Let X be the number of passengers who do not turn up for that flight.

(i) State, in context, two assumptions needed for X to be well modelled by a binomial distribution. [2]

(ii) Find the probability that fewer than 5 do not turn up for the flight. [2]

“Overbooking” occurs when there are more passengers who turn up than the number of seats available for the flight. When this happens, the airline will re-accommodate the excess passengers, that is, arrangements will be made for them to take another flight.

(iii) It is given that the particular flight has only 100 seats. Find the probability that at least 3 passengers have to be re-accommodated given that “overbooking” has occurred. [3]

17. **JJC/Prelim/2017/II/Q9**

A jar contains 5 identical balls numbered 1 to 5. A fixed number, n , of balls are selected and the number of balls with an even score is denoted by X .

- (i) Explain how the balls should be selected in order for X to be well modelled by a binomial distribution. [2]

Assume now that X has the distribution $B\left(n, \frac{2}{5}\right)$.

- (ii) Given that $n = 10$, find $P(X \geq 4)$. [2]
 (iii) Given that the mean of X is 4.8, find n . [2]
 (iv) Given that $P(X = 0 \text{ or } 1) < 0.01$, write down an inequality for n and find the least value of n . [3]

Shawn and Arvind take turns to draw one ball from the jar at random. The first person who draws a ball with an even score wins the game. Shawn draws first.

- (v) Show that the probability that Shawn wins the game is $\frac{3}{5}$ if the selection of balls is done without replacement. [2]
 (vi) Find the probability that Shawn wins the game if the selection of balls is done with replacement. [2]

18. **DHS 18/Prelim//II/Q10(i) – (ii)**

The discrete random variable X has probability mass function given by

$$P(X = x) = \begin{cases} \frac{ax}{n(n+1)} & \text{for } x = 1, 2, \dots, n, \\ 0 & \text{otherwise,} \end{cases}$$

where a is a constant.

- (i) Show that $a = 2$. [2]
 (ii) Find $E(X)$ in terms of n . You may use the result that $\sum_{r=1}^n r^2 = \frac{n}{6}(n+1)(2n+1)$. [3]

19. **MJC17/Prelim/II/Q6**

The probability function of X is given by

$$P(X = x) = \begin{cases} (2x-1)\theta & \text{if } x = 1, 2, 3 \\ k & \text{if } x = 4 \\ 0 & \text{otherwise} \end{cases}$$

where $0 < \theta < \frac{1}{9}$.

- (i) Show that $k = 1 - 9\theta$. Find, in terms of θ , the probability distribution of X . [2]
- (ii) Find $E(X)$ in terms of θ and hence show that $\text{Var}(X) = 26\theta - 196\theta^2$. [3]
- (iii) The random variable Y is related to X by the formula $Y = a + bX$, where a and b are non-zero constants. Given that $\text{Var}(Y) = \frac{1}{3}b^2$, find the value of θ . [3]

20. **AJC/Prelim/2018/II/Q6**

A farm produces a large number of eggs every day. The eggs are randomly packed into trays of 30. A small proportion p of these eggs are known to have weights which are substandard.

A check is carried out each day by taking a random sample of 50 trays and examining their weights. If more than 1 egg in a tray is substandard, the entire tray of eggs is rejected.

If 4 or more trays are rejected in the sample, the day's production is rated as 'poor'.

If none of the trays is rejected in the sample, the day's production is rated as 'excellent'.

Otherwise, the day's production is rated as 'fair'.

- (i) State, in this context, two conditions that must be met for the number of eggs in a tray that are substandard to be well modelled by a binomial distribution.

Assume now that these conditions are met.

- (ii) Show that the probability G that a tray is not rejected is given by

$$G = (1 - p)^{29} (1 + 29p).$$

Given that $G = 0.96$,

- (iii) find the value of p correct to four decimal places,
- (iv) find the probability that two trays are rejected if the day's production is rated as 'fair'.

21. **MI/BT/2017/II/Q5**

Ken participates in archery games, and the probability that he hits the bull's eye for each attempt is 0.73. For each game, he is given a set of 10 arrows. The number of times he hits the bull's eye in a game is denoted by X .

- (i) Calculate the probability that Ken hits the bull's eye at most 7 times in a game. [1]
- (ii) Find the probability that Ken hits the first bull's eye on his eighth arrow. [2]

Before the start of a particular game, Ken breaks his pair of glasses and the probability that he hits the bull's eye for each attempt is now p .

- (iii) He starts a new game and the probability of him hitting the bull's eye 8 times is now 0.12. Find the possible values of p . [2]
- State, with a reason, which value of p found is more appropriate in this case. [1]

22. **NJC/2019/BT2/II/6**

The number of rare words on a randomly chosen page in a book, X , can be modelled by the following probability distribution function.

$$P(X = x) = \begin{cases} 0.01 & \text{for } x = 0, \\ kx^{-2} & \text{for } 1 \leq x \leq 5, \\ 0 & \text{otherwise.} \end{cases}$$

- (i) Show that $k = 0.6764$, correct to 4 decimal places. [2]
- (ii) Find the expected value and variance of X . [2]

It can be assumed that rare words occur independently in the book.

- (iii) Find the probability that there are exactly a total of two rare words found on the first two pages. [3]
- (iv) State the probability that the first rare word is encountered on the n th page, leaving your answer in terms of n . [2]

23. SAJC/2019/BT2/II/9

Alex plays a game at a fun fair game stall, where he would remove balls at random from a bag containing 3 red balls and n green balls, one at a time, until a green ball is taken out. The number of red balls removed from the bag is denoted by R .

- (i) Find, in terms of n ,
 - (a) the probability distribution of R , [2]
 - (b) $E(R)$, [2]
 - (c) $E(R^2)$. [2]

You should simplify your answer as much as possible.

The amount of money Alex will be paid when playing the game is denoted by $\$X$, equal to the square of the number of red balls drawn.

- (ii) Find the expected amount of money Alex will be paid after playing the game once. [1]
- (iii) The owner of the fun-fair stall charges \$1 for playing the game once. Find the least number of green balls that the bag should contain such that the owner of the game stall expects to gain a profit after the first game. [3]
- (iv) Using your answer to (i)(b) and (i)(c), find the variance of R .

The owner of the game stall now decides to put a large number of green balls into the bag. Explain the meaning of the variance of R in the context of the question. [2]

24. **VJC/2019/BT2/12**

Café Dezerts produces ice cream cakes every day and discards what is not sold at the end of the day. The café earns \$15 for each ice cream cake sold and loses \$5 for each unsold ice cream cake. The demand per day for such ice cream cakes can be modelled by a random variable X with probability distribution as shown below.

x	0	1	2	3	4	5	6	7	8	≥ 9
$P(X = x)$	0.05	0.05	0.10	0.20	0.15	0.15	0.15	0.10	0.05	0

The café produces 8 ice cream cakes per day.

- (i) Find the probability that a total of 2 ice cream cakes are sold on two randomly chosen days. [2]
- (ii) Find $E(X)$ and $\text{Var}(X)$. [2]
- (iii) In a randomly chosen week, it is known that 3 ice cream cakes are sold per day on less than 5 days. Find the probability that 3 ice cream cakes are sold per day on at least 2 days in that week. [3]

The random variable Y is the total amount gained in dollars from the sales of ice cream cakes in a day.

- (iv) Show that $Y = 20X - 40$. [1]
- (v) Find the expected total amount gained in a day. [2]

The café is considering reducing the production of ice cream cakes to 6 per day. The random variable W is the number of ice cream cakes sold in a day when 6 ice cream cakes are produced. Assuming the same model for demand,

- (vi) explain why $P(W = 6) = P(X \geq 6)$, [1]
- (vii) show that the expected total amount gained in a day is \$50, [2]
- (viii) state, with a reason, if the café should consider the reduction in the number of ice cream cakes produced per day from 8 to 6. [1]

25. **IB/SL/2017/P2/Qn10**

The following table shows a probability distribution for the random variable X , where $E(X) = 1.2$.

x	0	1	2	3
$P(X = x)$	p	$\frac{1}{2}$	$\frac{3}{10}$	q

- (a) (i) Find the value of q .
(ii) Find the value of p .

A bag contains white and blue marbles, with at least three of each colour. Three marbles are drawn from the bag, without replacement. The number of blue marbles drawn is given by the random variable X .

- (b) (i) Write down the probability of drawing three blue marbles.
(ii) Explain why the probability of drawing three white marbles is $\frac{1}{6}$.
(iii) The bag contains a total of ten marbles of which w are white. Find the value of w .

A game is played in which three marbles are drawn from the bag of ten marbles, without replacement. A player wins a prize if three white marbles are drawn.

- (c) Jill plays the game nine times. Find the probability that she wins exactly two prizes.
(d) Grant plays the game until he wins two prizes. Find the probability that he wins his second prize on his eighth attempt.